ITA1441 – ETHICAL HACKING

LAB RECORD – EXPERIMENTS 1 TO 3

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EXPERIMENT 1: PORT SCANNING TOOLS

Experiment question: Perform port scanning using Nmap and observe the scan results.

Aim

To use Nmap to scan a host and identify the status of its TCP ports.

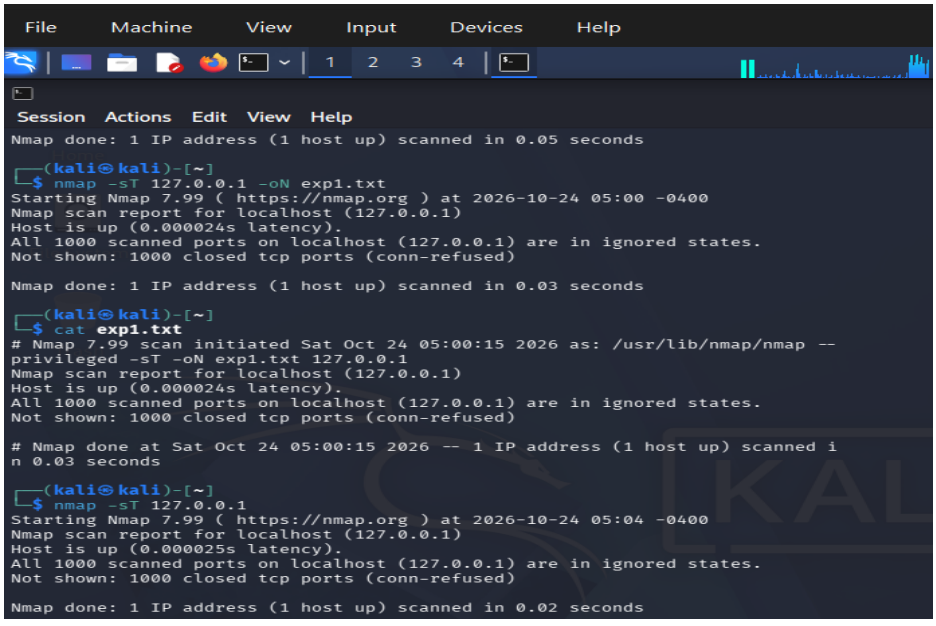
Algorithm

1. Open Kali Linux and launch Terminal.
2. Run an Nmap TCP connect scan against the local host (127.0.0.1).
3. Observe the host status and port information shown by Nmap.
4. Save or capture the terminal output for the lab record.

Command used

```
nmap -sT 127.0.0.1
```

Output Screenshot



Result

Nmap successfully scanned the local host. In the submitted output, the host was up and the scanned TCP ports were reported as closed/connection-refused.

EXPERIMENT 2: HOST DISCOVERY

Experiment question: Perform host discovery using Nmap.

Aim

To use Nmap to check whether a host is up without performing a port scan.

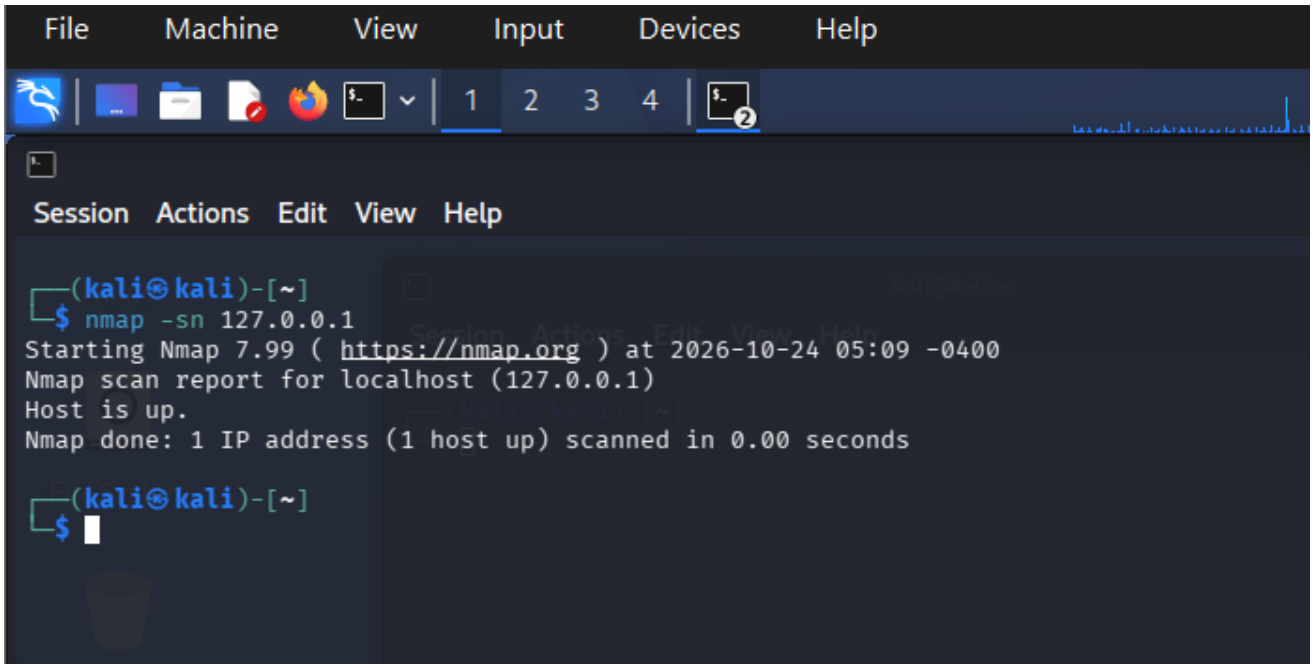
Algorithm

5. Open a fresh Kali Linux Terminal.
6. Run the Nmap host-discovery command for the local host.
7. Check the output for the host status.
8. Capture the command and output as evidence.

Command used

```
nmap -sn 127.0.0.1
```

Output Screenshot



Result

Nmap reported that localhost (127.0.0.1) was up and completed the host-discovery scan.

EXPERIMENT 3: CRACKING THE PASSWORD USING HYDRA

Experiment question: Explore Hydra, a tool used to test login/password combinations on authorized lab systems.

Aim

To identify Hydra and view its command-line options for password-testing tasks in an authorized practice environment.

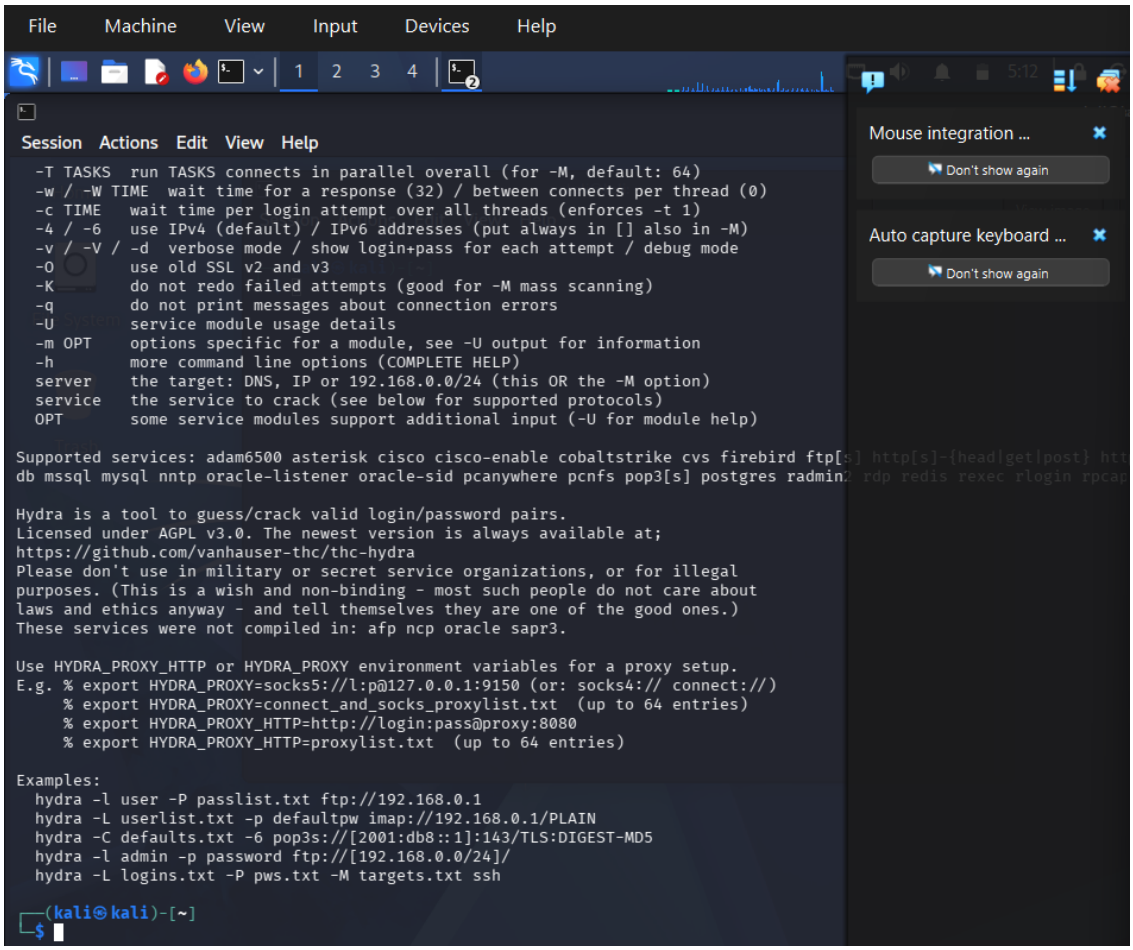
Algorithm

9. Open Kali Linux Terminal.
10. Run Hydra's help command to display its usage and supported options.
11. Review the help output and supported services.
12. Capture the help output for the lab record. Do not test accounts or systems without explicit authorization.

Command used

```
hydra -h
```

Output Screenshot



Result

Hydra displayed its command-line usage, options, and supported services. This confirms that the tool is available for learning in the authorized lab environment.